```
file_name <- "PRSA_Data_Aotizhongxin_20130301-20170228.csv"

air_data <- tryCatch(

{

    read.csv(file_name)

},

warning = function(w){

    message("Warning : ", w$message)

},

error = function(e){

    stop("Error : Unable to read the file.\nReason : ", e$message)

},

finally = {

    message("Import process completed.")

}

)
```

```
Import process completed.
```

```
if (!require(skimr)) install.packages("skimr")
if (!require(ggplot2)) install.packages("ggplot2")

library(skimr)
library(ggplot2)
```

```
cat("\nFirst Six Records\n")

head(air_data)
```

```
First Six Records
```

A data.frame: 6 × 18

| | No | year | month | day | hour | PM2.5 | PM10 | SO2 | NO2 | CO | O3 | TEMP | PRES | DEWP | RA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | <int> | <int> | <int> | <int> | <int> | <dbl> | <dbl> | <dbl> | <dbl> | <int> | <dbl> | <dbl> | <dbl> | <dbl> | <db |
| **1** | 1 | 2013 | 3 | 1 | 0 | 4 | 4 | 4 | 7 | 300 | 77 | -0.7 | 1023.0 | -18.8 | |
| **2** | 2 | 2013 | 3 | 1 | 1 | 8 | 8 | 4 | 7 | 300 | 77 | -1.1 | 1023.2 | -18.2 | |
| **3** | 3 | 2013 | 3 | 1 | 2 | 7 | 7 | 5 | 10 | 300 | 73 | -1.1 | 1023.5 | -18.2 | |
| **4** | 4 | 2013 | 3 | 1 | 3 | 6 | 6 | 11 | 11 | 300 | 72 | -1.4 | 1024.5 | -19.4 | |
| **5** | 5 | 2013 | 3 | 1 | 4 | 3 | 3 | 12 | 12 | 300 | 72 | -2.0 | 1025.2 | -19.5 | |
| **6** | 6 | 2013 | 3 | 1 | 5 | 5 | 5 | 18 | 18 | 400 | 66 | -2.2 | 1025.6 | -19.6 | |

```
cat("\nStructure of Dataset\n")

str(air_data)
```

```
Structure of Dataset
'data.frame':   35064 obs. of  18 variables:
 $ No     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ year   : int  2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
 $ month  : int  3 3 3 3 3 3 3 3 3 3 ...
 $ day    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ hour   : int  0 1 2 3 4 5 6 7 8 9 ...
 $ PM2.5  : num  4 8 7 6 3 5 3 3 3 3 ...
 $ PM10   : num  4 8 7 6 3 5 3 6 6 8 ...
 $ SO2    : num  4 4 5 11 12 18 18 19 16 12 ...
 $ NO2    : num  7 7 10 11 12 18 32 41 43 28 ...
 $ CO     : int  300 300 300 300 300 400 500 500 500 400 ...
 $ O3     : num  77 77 73 72 72 66 50 43 45 59 ...
 $ TEMP   : num  -0.7 -1.1 -1.1 -1.4 -2 -2.2 -2.6 -1.6 0.1 1.2 ...
 $ PRES   : num  1023 1023 1024 1024 1025 ...
 $ DEWP   : num  -18.8 -18.2 -18.2 -19.4 -19.5 -19.6 -19.1 -19.1 -19.2 -19.3 ...
 $ RAIN   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ wd     : chr  "NNW" "N" "NNW" "NW" ...
 $ WSPM   : num  4.4 4.7 5.6 3.1 2 3.7 2.5 3.8 4.1 2.6 ...
 $ station: chr  "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" ...
```

```
cat("\nRows :", nrow(air_data))

cat("\nColumns :", ncol(air_data))
```

```
Rows : 35064
Columns : 18
```

```
cat("\nDoes dataset contain missing values?\n")

any(is.na(air_data))
```

```
Does dataset contain missing values?
TRUE
```

```
cat("\nTotal Missing Values : ")

sum(is.na(air_data))
```

```
Total Missing Values :
7271
```

```
# Import dataset
air_data <- read.csv(file_name)

# Save original dataset BEFORE cleaning
original_air_data <- air_data
```

```
temperature <- c(28, 30, NA, 32)

temperature
```

```
28 · 30 · <NA> · 32
```

```
cat("\nChecking NA Values\n")

is.na(temperature)
```

```
Checking NA Values
FALSE · FALSE · TRUE · FALSE
```

```
missing_object <- NULL

missing_object
```

```
NULL
```

```
cat("\nChecking NULL Object\n")

is.null(missing_object)
```

```
Checking NULL Object
TRUE
```

```
undefined_value <- 0 / 0

undefined_value
```

```
NaN
```

```
cat("\nChecking NaN Value\n")

is.nan(undefined_value)
```

```
Checking NaN Value
TRUE
```

```
cat("\nNA Check\n")
print(is.na(NA))
print(is.null(NA))
print(is.nan(NA))
```

```
NA Check
[1] TRUE
[1] FALSE
[1] FALSE
```

```
cat("\nNULL Check\n")
print(is.na(NULL))
print(is.null(NULL))
print(is.nan(NULL))
```

```
NULL Check
logical(0)
[1] TRUE
logical(0)
```

```
cat("\nNaN Check\n")
print(is.na(NaN))
print(is.null(NaN))
print(is.nan(NaN))
```

```
print(is.nan(NaN))
```

```
NaN Check
[1] TRUE
[1] FALSE
[1] TRUE
```

```
temperature <- c(28, 30, NA, 32)

cat("\nTemperature Vector:\n")
print(temperature)

cat("\nChecking NA:\n")
print(is.na(temperature))

missing_object <- NULL

cat("\nNULL Object:\n")
print(missing_object)

cat("\nChecking NULL:\n")
print(is.null(missing_object))

undefined_value <- 0 / 0

cat("\nNaN Value:\n")
print(undefined_value)

cat("\nChecking NaN:\n")
print(is.nan(undefined_value))



cat("\nNA Checks\n")
print(is.na(NA))
print(is.null(NA))
print(is.nan(NA))

cat("\nNULL Checks\n")
print(is.na(NULL))
print(is.null(NULL))
print(is.nan(NULL))

cat("\nNaN Checks\n")
print(is.na(NaN))
print(is.null(NaN))
print(is.nan(NaN))
```

```
Temperature Vector:
[1] 28 30 NA 32

Checking NA:
[1] FALSE FALSE  TRUE FALSE

NULL Object:
NULL

Checking NULL:
[1] TRUE

NaN Value:
[1] NaN
```

```
Checking NaN:
[1] TRUE

NA Checks
[1] TRUE
[1] FALSE
[1] FALSE

NULL Checks
logical(0)
[1] TRUE
logical(0)

NaN Checks
[1] TRUE
[1] FALSE
[1] TRUE
```

```
selected_variables <- c(
  "PM2.5",
  "PM10",
  "SO2",
  "NO2",
  "TEMP",
  "WSPM",
  "wd"
)
```

```
missing_summary <- function(data)
{

    summary_table <- data.frame(
        Variable = character(),
        Total_Records = integer(),
        Missing_Values = integer(),
        Missing_Percentage = numeric(),
        stringsAsFactors = FALSE
    )

    for(variable in selected_variables)
    {
        if(variable %in% names(data))
        {

            total_records <- nrow(data)

            missing_values <- sum(is.na(data[[variable]]))

            missing_percentage <- (missing_values / total_records) * 100

            summary_table <- rbind(
                summary_table,

                data.frame(

                    Variable = variable,

                    Total_Records = total_records,

                    Missing_Values = missing_values,

                    Missing_Percentage = round(missing_percentage,2)
```

```
                )

            )


            if(missing_percentage > 20)
            {

                warning(
                    paste(
                        variable,
                        "contains more than 20% missing values."
                    )
                )

            }

        }

        else
        {

            warning(
                paste(
                    variable,
                    "does not exist in dataset."
                )
            )

        }

    }

    return(summary_table)

}
```

```
summary_result <- missing_summary(air_data)

cat("\nMissing Value Summary\n")

print(summary_result)
```

```
Missing Value Summary
  Variable Total_Records Missing_Values Missing_Percentage
1    PM2.5         35064            925               2.64
2     PM10         35064            718               2.05
3      SO2         35064            935               2.67
4      NO2         35064           1023               2.92
5     TEMP         35064             20               0.06
6     WSPM         35064             14               0.04
7       wd         35064             81               0.23
```

```
air_data$pollution_ratio <- air_data$PM2.5 / air_data$PM10
```

```
na_count <- sum(is.na(air_data$pollution_ratio))
```

```
nan_count <- sum(is.nan(air_data$pollution_ratio))
```

```
infinite_count <- sum(is.infinite(air_data$pollution_ratio))
```

```
positive_inf <- sum(air_data$pollution_ratio == Inf, na.rm = TRUE)
```

```
negative_inf <- sum(air_data$pollution_ratio == -Inf, na.rm = TRUE)
```

```
cat("\nPollution Ratio Summary\n")

cat("NA Values :", na_count, "\n")

cat("NaN Values :", nan_count, "\n")

cat("Positive Infinity :", positive_inf, "\n")

cat("Negative Infinity :", negative_inf, "\n")

cat("Total Infinite Values :", infinite_count, "\n")
```

```
Pollution Ratio Summary
NA Values : 934
NaN Values : 0
Positive Infinity : 0
Negative Infinity : 0
Total Infinite Values : 0
```

```
air_data$pollution_ratio[
    is.nan(air_data$pollution_ratio)
] <- NA

air_data$pollution_ratio[
    is.infinite(air_data$pollution_ratio)
] <- NA
```

```
cat("\nAfter Cleaning\n")

cat("NaN :", sum(is.nan(air_data$pollution_ratio)), "\n")

cat("Infinite :", sum(is.infinite(air_data$pollution_ratio)), "\n")

cat("NA :", sum(is.na(air_data$pollution_ratio)), "\n")
```

```
After Cleaning
NaN : 0
Infinite : 0
NA : 934
```

```
#----------------------------------------
# Task 5 : Handle Missing Numerical Values
#----------------------------------------

numeric_variables <- c(
  "PM2.5",
  "PM10",
  "SO2",
  "NO2",
  "TEMP",
  "WSPM"
)
```

```
for(variable in numeric_variables)
{

    # Check whether column exists
    if(variable %in% names(air_data))
    {

        # Missing values before cleaning
        missing_before <- sum(is.na(air_data[[variable]]))

        # Median calculation
        median_value <- median(
            air_data[[variable]],
            na.rm = TRUE
        )

        # Replace missing values
        air_data[[variable]][
            is.na(air_data[[variable]])
        ] <- median_value

        # Missing values after cleaning
        missing_after <- sum(is.na(air_data[[variable]]))

        # Display information
        cat("\n---------------------------------\n")

        cat("Variable :", variable, "\n")

        cat("Missing Before :", missing_before, "\n")

        cat("Median Used :", median_value, "\n")

        cat("Missing After :", missing_after, "\n")

    }

    else
    {

        cat("\nVariable", variable, "does not exist.\n")

    }

}
```

```
---------------------------------
Variable : PM2.5
Missing Before : 925
Median Used : 58
Missing After : 0

---------------------------------
Variable : PM10
Missing Before : 718
Median Used : 87
Missing After : 0

---------------------------------
Variable : SO2
Missing Before : 935
Median Used : 9
Missing After : 0
```

```
---------------------------------
Variable : NO2
Missing Before : 1023
Median Used : 53
Missing After : 0

---------------------------------
Variable : TEMP
Missing Before : 20
Median Used : 14.5
Missing After : 0

---------------------------------
Variable : WSPM
Missing Before : 14
Median Used : 1.4
Missing After : 0
```

```
#---------------------------------------
# Task 6 : Handle Missing Categorical Values
#---------------------------------------

calculate_mode <- function(x)
{
    # Remove missing values
    x <- x[!is.na(x)]

    # Check if all values are missing
    if(length(x) == 0)
    {
        return(NA)
    }

    # Frequency table
    frequency_table <- table(x)

    # Most frequent value
    mode_value <- names(which.max(frequency_table))

    return(mode_value)
}

# Missing values before replacement
missing_before <- sum(is.na(air_data$wd))

# Calculate mode
mode_value <- calculate_mode(air_data$wd)

# Replace missing values
air_data$wd[
    is.na(air_data$wd)
] <- mode_value

# Missing values after replacement
missing_after <- sum(is.na(air_data$wd))

# Display results
cat("\nWind Direction Cleaning\n")

cat("Missing Before :", missing_before, "\n")

cat("Mode Used :", mode_value, "\n")

cat("Missing After :", missing_after, "\n")
```

```
Wind Direction Cleaning
Missing Before : 81
Mode Used : NE
Missing After : 0
```

```
#---------------------------------------
# Task 7 : Implement Error Handling
#---------------------------------------

# User-defined function to clean a numerical variable
clean_variable <- function(data, variable_name)
{
  tryCatch({

    # Check if the variable exists
    if (!(variable_name %in% names(data))) {
      stop(paste("Variable", variable_name, "does not exist."))
    }

    # Check if the variable is numerical
    if (!is.numeric(data[[variable_name]])) {
      stop(paste("Variable", variable_name, "is not numerical."))
    }

    # Check if all values are missing
    if (all(is.na(data[[variable_name]]))) {
      stop(paste("Variable", variable_name, "contains only missing values."))
    }

    # Calculate the median
    median_value <- median(data[[variable_name]], na.rm = TRUE)

    # Check if median could not be calculated
    if (is.na(median_value)) {
      stop(paste("Median could not be calculated for", variable_name))
    }

    # Count missing values before replacement
    missing_before <- sum(is.na(data[[variable_name]]))

    # Replace missing values with median
    data[[variable_name]][is.na(data[[variable_name]])] <- median_value

    # Count missing values after replacement
    missing_after <- sum(is.na(data[[variable_name]]))

    # Display results
    cat("\n---------------------------------\n")
    cat("Variable :", variable_name, "\n")
    cat("Missing Before :", missing_before, "\n")
    cat("Median Used :", median_value, "\n")
    cat("Missing After :", missing_after, "\n")
    cat(variable_name, "cleaned successfully.\n")

    # Return cleaned variable
    return(data[[variable_name]])

  },

  error = function(e) {

    message("Error: ", e$message)
```

```
    return(NULL)

  })
}

#----------------------------------------
# Testing the Function
#----------------------------------------

# Successful cleaning
clean_pm25 <- clean_variable(air_data, "PM2.5")
clean_pm10 <- clean_variable(air_data, "PM10")
clean_temp <- clean_variable(air_data, "TEMP")

# Test: Variable does not exist
clean_variable(air_data, "ABC")

# Test: Categorical variable
clean_variable(air_data, "wd")

# Test: Variable with all missing values
air_data$dummy <- NA
clean_variable(air_data, "dummy")
```

```
----------------------------------
Variable : PM2.5
Missing Before : 0
Median Used : 58
Missing After : 0
PM2.5 cleaned successfully.

----------------------------------
Variable : PM10
Missing Before : 0
Median Used : 87
Missing After : 0
PM10 cleaned successfully.

----------------------------------
Variable : TEMP
Missing Before : 0
Median Used : 14.5
Missing After : 0
TEMP cleaned successfully.
Error: Variable ABC does not exist.

NULL
Error: Variable wd is not numerical.

NULL
Error: Variable dummy is not numerical.

NULL
```

```
#----------------------------------------
# Task 8 : Compare Missing Values Before and After Cleaning
#----------------------------------------

# Variables to compare
comparison_variables <- c(
  "PM2.5",
  "PM10",
  "SO2",
  "NO2",
  "TEMP",
```

```
  "WSPM",
  "wd"
)

# Create empty comparison table
comparison_table <- data.frame(
  Variable = character(),
  Missing_Before = integer(),
  Missing_After = integer(),
  Values_Replaced = integer(),
  stringsAsFactors = FALSE
)

# Loop through each variable
for(variable in comparison_variables)
{

  # Missing values before cleaning
  before <- sum(is.na(original_air_data[[variable]]))

  # Missing values after cleaning
  after <- sum(is.na(air_data[[variable]]))

  # Values replaced
  replaced <- before - after

  # Add row to comparison table
  comparison_table <- rbind(
    comparison_table,
    data.frame(
      Variable = variable,
      Missing_Before = before,
      Missing_After = after,
      Values_Replaced = replaced
    )
  )

}

# Display comparison table
cat("\nComparison of Missing Values Before and After Cleaning\n\n")

print(comparison_table)
```

```
Comparison of Missing Values Before and After Cleaning

  Variable Missing_Before Missing_After Values_Replaced
1    PM2.5            925             0              925
2     PM10            718             0              718
3      SO2            935             0              935
4      NO2           1023             0             1023
5     TEMP             20             0               20
6     WSPM             14             0               14
7       wd             81             0               81
```

```
library(ggplot2)
```

```
#----------------------------------------
# Task 9 : Visualization of Missing Values
#----------------------------------------

plot_data <- data.frame(
```

```
  Variable = rep(comparison_table$Variable, 2),

  Missing_Values = c(
    comparison_table$Missing_Before,
    comparison_table$Missing_After
  ),

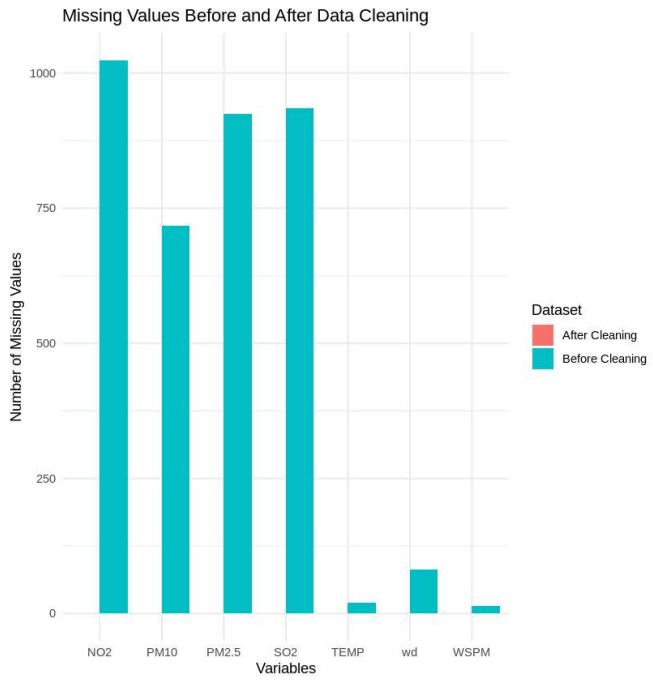
  Status = c(
    rep("Before Cleaning", nrow(comparison_table)),
    rep("After Cleaning", nrow(comparison_table))
  )
)
```

```
ggplot(plot_data,
       aes(x = Variable,
           y = Missing_Values,
           fill = Status)) +

  geom_bar(stat = "identity",
           position = "dodge") +

  labs(
    title = "Missing Values Before and After Data Cleaning",
    x = "Variables",
    y = "Number of Missing Values",
    fill = "Dataset"
  ) +

  theme_minimal()
```




```
#---------------------------------------
# Task 10 : Export the Cleaned Dataset
#---------------------------------------

tryCatch({

  # Export cleaned dataset
  write.csv(
    air_data,
```

```
    file = "cleaned_air_quality_data.csv",
    row.names = FALSE
  )

  cat("\n----------------------------------------\n")
  cat("Cleaned dataset exported successfully.\n")
  cat("File Name : cleaned_air_quality_data.csv\n")
  cat("----------------------------------------\n")

},

warning = function(w){

  message("Warning: ", w$message)

},

error = function(e){

  message("Error while exporting file: ", e$message)

},

finally = {

  cat("\nExport process completed.\n")

})
```

```
----------------------------------------
Cleaned dataset exported successfully.
File Name : cleaned_air_quality_data.csv
----------------------------------------

Export process completed.
```